

No-Code Dataset Preparation Platform

A No Code approach towards data cleaning and preprocessing

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Requirements Analysis

1. Introduction

The system is a no-code dataset cleaning and preprocessing web application designed to prepare raw datasets for machine learning and data analysis. The system allows users to upload datasets and perform common data cleaning and preprocessing operations through a graphical user interface without requiring programming knowledge.

The requirements analysis identifies the needs of the intended users and defines the capabilities and behaviour of the proposed system.

The requirements are derived from the primary objective of simplifying the dataset preprocessing workflow while providing users with control over the transformations applied to their data.

The system is intended to support common dataset problems such as missing values, duplicate records, incorrect data types, inconsistent entries, outliers, redundant features, formatting inconsistencies, categorical data and normalization.

2. Overall System Requirements

The system shall provide a web-based interface through which a user can upload a dataset, inspect its characteristics, apply preprocessing operations, review the resulting dataset, and export the processed dataset.

The system should:

1. Allow users to upload supported dataset files such as CSV and XLSX.
2. Read the uploaded dataset and represent it internally for processing.
3. Display an initial preview and summary of the dataset.
4. Provide preprocessing operations through graphical controls.
5. Allow users to select appropriate columns and preprocessing methods.
6. Apply selected operations to the dataset.
7. Maintain the updated dataset throughout the preprocessing workflow.
8. Display the results of preprocessing operations.
9. Maintain a history or log of operations performed by the user.
10. Allow users to inspect the processed dataset before export.
11. Allow users to download the final processed dataset.
12. Prevent or handle invalid preprocessing operations where possible.
13. Provide meaningful feedback when an operation succeeds or fails.
14. Provide a simple interface that does not require the user to write code.

3. Dataset Requirements

The system shall support datasets containing numerical, categorical, textual, and other commonly encountered data types.

The preprocessing workflow should be capable of handling datasets containing:

- Missing or null values
- Duplicate records
- Incorrect data types
- Inconsistent values
- Invalid values
- Outliers
- Redundant columns
- Highly correlated features
- Mixed formatting
- Inconsistent casing and whitespace
- Categorical variables
- Features requiring scaling or normalization

The system should preserve the relationship between rows and columns while applying transformations and should ensure that the resulting dataset remains suitable for subsequent machine learning workflows.

4. Preprocessing Requirements

The system shall provide separate preprocessing operations for different types of dataset problems.

4.1 Missing Values

The system shall allow users to handle missing values using methods such as:

- Mean
- Median
- Mode
- Constant value
- Forward fill
- Backward fill

The user should be able to select the columns on which the operation is to be performed.

4.2 Duplicate Records

The system shall identify and allow users to remove duplicate records from the dataset.

4.3 Incorrect Data Types

The system shall allow users to convert columns into appropriate data types, such as numerical, categorical, or textual types.

The system should also identify values that cannot be converted and provide an appropriate handling mechanism.

4.4 Inconsistent Entries

The system shall allow inconsistent values within a column to be standardized or converted into a specified value.

4.5 Outliers

The system shall provide methods for identifying and handling numerical outliers.

Possible techniques include:

- Z-score
- Interquartile Range (IQR)

The system shall allow identified outliers to be removed or otherwise handled according to the selected operation.

4.6 Invalid Data

The system shall provide mechanisms for identifying and handling invalid values where applicable.

4.7 Highly Correlated Features

The system should identify highly correlated numerical features and allow redundant correlated features to be removed while retaining an appropriate representative feature.

4.8 Redundant Columns

The system shall allow users to remove unnecessary or redundant columns from the dataset.

4.9 Formatting

The system shall provide operations for standardizing inconsistent formatting within dataset values.

4.10 Case and Whitespace

The system shall provide operations for handling inconsistent capitalization and unnecessary leading or trailing whitespace in textual data.

4.11 Encoding

The system shall provide encoding techniques for categorical variables, including:

- One-Hot Encoding
- Ordinal Encoding
- Binary Encoding
- Target Encoding where appropriate

4.12 Feature Scaling

The system shall provide scaling methods such as:

- Min-Max Scaling
- Robust Scaling

4.13 Normalization

The system shall provide normalization/standardization functionality for numerical features where required by the selected machine learning workflow.

4.14 Feature Selection

The system should provide mechanisms for selecting relevant features and reducing unnecessary features from the dataset.

5. Processing and Workflow Requirements

The system shall follow a sequential preprocessing workflow.

The general workflow shall be:

Upload Dataset -> Inspect Dataset -> Identify Data Issues -> Select Preprocessing Operation -> Configure Operation -> Apply Operation -> Review Result -> Repeat Operations -> Export Dataset

The system shall maintain the latest version of the dataset while the user performs multiple preprocessing operations.

Each operation should be independent and should have a clearly defined purpose. The system should record the operation performed, the selected method, and the relevant columns or parameters where applicable.

6. User Interface Requirements

The application shall provide a graphical interface suitable for users with limited or no programming knowledge.

The interface should provide:

- Dataset upload control
- Dataset preview
- Dataset summary
- Preprocessing operation controls
- Column selection controls
- Method selection controls
- Parameter input fields where required
- Operation status messages
- Transformation log
- Processed dataset preview
- Dataset download functionality

Preprocessing operations should be organized in a structured manner so that users can easily identify the appropriate operation for a particular data problem.

7. Validation Requirements

The system shall validate user inputs before applying preprocessing operations.

Examples include:

- Ensuring that at least one column is selected when required.
- Preventing numerical operations from being applied to incompatible textual columns.
- Validating user-provided constant values.
- Checking whether required data is available before applying an operation.
- Providing an understandable error message when an operation cannot be performed.

The system should avoid silently producing incorrect transformations.

8. Output Requirements

After preprocessing, the system shall allow the user to inspect the resulting dataset.

The system shall provide:

1. Updated dataset preview.
2. Updated dataset summary.
3. Information about transformations performed.
4. Final processed dataset export.

The system should support exporting the processed dataset in supported formats such as CSV and XLSX.

9. Logging Requirements

The system should maintain a log of preprocessing operations performed during the current session.

The log should provide information such as:

- Operation performed
- Method used
- Columns affected
- Relevant parameters

This will allow users to understand how the raw dataset was transformed into the final dataset.

10. Performance Requirements

The system should process datasets efficiently while maintaining acceptable response times. For the initial prototype, the application may primarily operate on datasets that can reasonably be loaded into memory.

The architecture should, however, be designed so that future versions can support significantly larger datasets using technologies such as columnar storage and query engines without requiring a complete redesign of the preprocessing workflow. This will be handled under scalability.

The system should avoid unnecessary duplication of the dataset during preprocessing operations wherever possible.

11. Security and Reliability Requirements

The system should:

- Validate uploaded files before processing.
- Reject unsupported file formats.
- Handle malformed datasets gracefully.
- Prevent application crashes caused by invalid user input.
- Avoid exposing dataset contents unnecessarily.
- Maintain data consistency during preprocessing.
- Provide meaningful error messages.
- Ensure that the exported dataset reflects the operations successfully applied by the user.

12. Extensibility Requirements

The system should be designed so that additional preprocessing techniques can be added without requiring major changes to the main application.

Each preprocessing operation should ideally be implemented as an independent module with:

- Operation name/title
- Supported column/data type
- Available methods
- Required parameters
- Processing function
- Result/log information

This modular design will allow additional preprocessing techniques to be incorporated in future versions.

13. Assumptions

The following assumptions are made during requirements analysis:

1. The primary user has basic knowledge of datasets and machine learning concepts.
2. The user does not need to know Python or write preprocessing code.
3. The uploaded dataset contains structured tabular data and is within the specified boundaries of the file type.
4. The user is familiar with the specific dataset they are dealing with and have studied it to their best capability.
5. The user is responsible for selecting preprocessing methods appropriate for their dataset.
6. The initial version focuses on common preprocessing operations rather than every possible data-cleaning technique.
7. The initial prototype operates primarily on datasets that can be processed within available system memory.
8. Machine learning model training is outside the primary scope of the system.

14. Constraints

The project has the following constraints:

- The initial application is intended to be a lightweight web application.
- The prototype is primarily designed for tabular datasets.
- Processing very large datasets may be limited by available memory.
- Some preprocessing techniques are dependent on the data type and structure of the dataset.
- Certain operations may require users to understand the meaning of their data before applying them.
- The application focuses on preprocessing and does not replace domain-specific data validation.
- The initial implementation may not support every possible file format or preprocessing algorithm.